Structural and Reduced Form Models
ECON 8120
Spring 2026

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Introduction

- Outline for the first few classes....

  1. Overview of structural and reduced form methods
  2. Approach to writing a research paper that uses a structural model
  3. Non-linear estimation and computation

CAUSALITY: THE GOAL OF ECONOMETRICS

- In almost all econometric endeavors, the goal is to uncover causal relationships (as opposed to prediction)
- Causality is crucial because it's the only way to confidently make counterfactual predictions
  - note that economic theory makes causal predictions: demand is downward sloping
  - another example, what is the effect of reading to my child on her cognitive development?
  - we can't answer this by simply looking at $Corr(\text{read to child}, \text{cog. test score})$
  - this correlation is contaminated by omitted variable bias:
    - parents who regularly read to their children likely also regularly do other things favorable to child development

CAUSALITY REQUIRES A COUNTERFACTUAL

- Causality is defined in terms of a *counterfactual*
  - what would've been the outcome if everything were the same absent a policy/choice/event?
  - this is the notion of *ceteris paribus* in principles of economics
  - theory says quantity demanded will decrease if price increases, *ceteris paribus*
  - what is child's test score if everything were the same *except* regular reading time?
- "Causal effect": difference between reality and the most plausible counterfactual

TWO BROAD EMPIRICAL APPROACHES

1. **Descriptive**
   - Document facts about observable quantities
   - These facts sometimes suggest a causal relationship
   - e.g. raw earnings difference between college graduates and HS graduates

2. **Structural**
   - Estimate parameters of a data generating process (DGP) which are assumed to be invariant to policy changes or other counterfactuals
   - Once we know the DGP, we can make causal inferences
     - e.g. estimate a DGP that specifies how cognitive ability and family background relate to the decision to enroll in college and to post-graduation earnings

BRIEF HISTORY OF THE TERM "STRUCTURAL"

- Dates back to Hurwicz (1950) and Koopmans and Reiersol (1950)
- A "structure" is a data generating process
  - i.e. a set of functional or probabilistic relationships between observable and latent variables which implies a joint distribution of the observables
- The goal of structural estimation, then, is to estimate the parameters (or functions) of the DGP
- This allows us to make counterfactual comparisons, i.e. perform causal inference
- Note that "structural" here refers to basically all of modern econometrics

BRIEF HISTORY OF THE TERM "REDUCED FORM"

- The term "reduced form" refers to solving a structural model
- The structural model may have endogenous variables on both sides of the equation
- But the reduced form puts all endogenous variables on the left hand side
- All exogenous variables and error terms are on the right hand side
- Classic example: supply and demand
  - Two equations, two endogenous variables (price, quantity)
  - In equilibrium, reduced form has P and Q as respective LHS variables
  - RHS contains observable and unobservable determinants of S and D

HOW THESE TERMS TEND TO BE USED TODAY

- Reduced form tends to refer to linear models estimated by RCT, IV, DID, RDD, etc.
  - Methods that try to exploit randomization (or quasi-randomization)
  - Synonymous with the phrase "identification strategy"
  - Goal is typically estimation of a "treatment" effect

- Structural tends to refer to non-linear models aimed at estimating economic primitives that are policy-invariant
  - often make explicit a set of maintained assumptions
  - focus on settings where RCTs would be infeasible

- Both of these terms are misnomers, but this is how they are used today

- For the rest of these slides, we will similarly misuse them

Structural and Reduced Form Methods

- There is often friction between structural and reduced-form practitioners
- Keane (2010) calls reduced-form methods "atheoretic"
  - others refer to reduced-form approaches as "cuteonomics"
  - why? RF methods often provide little understanding of mechanisms or the impact of alternative policies
- Angrist and Pischke (2009) titled their book "Mostly Harmless Econometrics"
  - implying that structural methods include "harmful" econometrics (Lewbel, 2019)
  - why? identification is often less clear in structural models and they rely heavily on functional form, distributional, and behavioral assumptions
- Of course both methodological approaches can help answer interesting policy questions (and can be combined quite succuessfully)

Finding the right approach

- Blundell (2010) outlines three broad concerns determining what method to use:
    1. The nature of the question to be answered
    2. The type and quality of data available
    3. The mechanism by which individuals are allocated or receive the policy
        - i.e. random assignment or obeying rules of economic theory (like utility or profit maximization)
- "Just as an experiment needs to be carefully designed, the identification of a structural economic model needs to be carefully argued."
- "Poorly designed quasi-experiments have little to offer, but so too do poorly focused structural estimations"

SCARCITY OF STRUCTURAL PAPERS

- Reduced form methods are more commonly applied - especially in certain applied micro fields
- This is mainly because entry barriers are low and the implementation of structural methods can be difficult
- For every structural paper produced, 3 or 4 RF papers could be produced
- This matters for publishing, tenure, and training the next generation
- Nonetheless, structural methods can be immensely useful
- But they take longer, so their science progresses relatively more slowly

RCT as structural estimation

- All causal inference is structural in nature (as correctly defined)
- Structural estimation need not be difficult; models need not be complex
- An RCT is a structural model that can be evaluated descriptively
  - No fancy econometrics needed: just compute $\overline{y}_{\text{Treatment}} - \overline{y}_{\text{Control}}$
- This is because great effort was expended at the randomization step
- The experimenters had a (structural) model in mind when defining treatment
  - e.g. Berry & Dizon-Ross (2021)

Not every DGP can be evaluated by RCT

- Some causal questions of interest can't be answered with an RCT
- For a variety of reasons, experimentation may be too costly, unethical, etc.
  - e.g. can't randomize a merger of two large firms, or a person's height
- Without explicit randomization, we have to rely on observational data
- This requires more complex econometric methods to estimate the DGP
  - Need to explicitly specify how unobservables relate to other parts of model

Key parameters of any economic model

- As mentioned above, we assume that DGP parameters are policy-invariant
- These parameters tend to be related to economic fundamentals:
  - preferences
  - production technology
  - commodities
  - demographics
  - beliefs and expectations
  - space (includes networks & social interactions)
- What is typically not policy invariant are choices.....they change as we alter prices, legislation, information, etc.

Reading-to-Children: Reduced-form example

- A "reduced-form" (as misused today) approach would look like the following:

  1. recruit a group of families to participate in a reading study
  2. randomize into "no-read" and "read" groups
  3. after some period of time, give their children a cognitive test
  4. compare the average scores of children across each of the groups

READING-TO-CHILDREN: STRUCTURAL APPROACH

- A "structural" (as misused today) approach would look like the following:

  1. write a model of child skill formation, Cunha et al. (2010)
  2. gather data on parental investments, income, time-use, and child test scores
  3. estimate the parameters of the child skill formation model
  4. use model to simulate counterfactual policies (e.g. where reading is set to 0)
  5. compare average scores in counterfactual and status quo

Reading-to-Children: Hybrid approach

- A hybrid approach might do the following:

  1. estimate the skill formation parameters
  2. leverage randomization to better estimate/validate the model
     - e.g. by allowing for identification of a parameter previously not identifiable
     - e.g. recover randomization-implied ATE using structural parameter estimates
  3. use the validated structural model to explore other counterfactuals

- A great example of this hybrid approach is Delavande and Zafar (2019)

WHAT IS IDENTIFICATION?

- Identification is a critical element for both structural and reduced form methods
- A model is identified when parameters are *uniquely determined* from the observable population generating the data
- identification is never a question about a sample of data
- it is a question about the population from which the sample is drawn
- there are many different terms for identification in econometrics but the unifying definition is the one above
- Lewbel (2019) lists 33 different terms from the econometrics literature

Why is identification important?

- The study of identification logically precedes estimation, inference, and testing
- For θ to be identified, alternative values of θ must imply different distributions of the observable data
- If θ is not identified, then we cannot hope to find a consistent estimator for θ
- More generally, identification failures complicate statistical analyses of models, so recognizing lack of identification and searching for restrictions that suffice to attain identification are fundamentally important problems in econometric modeling

REDUCED-FORM VS. STRUCTURAL IDENTIFICATION

- How is "identification" used differently in reduced-form vs. structural econometrics?
- In reduced-form econometrics (a.k.a. causal modeling):
  - Typically talk of an "identification strategy" (i.e. randomization setup)
  - Focus is on estimation of treatment effects, not the DGP parameters
  - Relies on randomization from some kind of randomized or natural experiment
- In structural econometrics:
  - Typically talk of "establishing identification" (i.e. sufficient variation in data)
  - In complex models, can be difficult to do without imposing more assumptions

THE CREDIBILITY REVOLUTION

- What makes an identification strategy credible?
- Identification means separating selection from treatment
- This is best done when treatment is randomized
- Randomization is also how the natural sciences make discoveries
  - Typically done via controlled experiments
- The closer a reduced-form model is to an RCT, the better
  - note that controlled experiments are impossible to do with humans

Examples of RF Identification Strategies

- Randomized experiments, field experiments, lab experiments
- Instrumental variables, regression discontinuity
- Difference in differences, synthetic control methods
- Matching methods (nearest neighbor, propensity score, ...)
- OLS that does not suffer from omitted variable bias
- These are almost exclusively estimated using linear econometric models
- Credibility is proportional to the plausibility of randomization
  - often hear whether something is "cleanly" identified

SOURCES OF IDENTIFICATION - STRUCTURAL

- Exogenous variation in observables
  - prices, wages, policies, demographics
  - analogous to instruments in linear models
- Functional form and distributional assumptions
  - parametric utility or production functions
  - assumptions on unobserved shocks (e.g. EV1, normal)
- Behavioral assumptions
  - optimization (utility or profit maximization)
  - rational expectations, forward-looking behavior
- Dynamic restrictions
  - state transitions and law of motion
  - exclusion restrictions across time
- Cross-equation restrictions
  - joint determination of choices and outcomes
  - equilibrium conditions

CREDIBLE STRUCTURAL MODELS

- What makes a structural model credible?
- At the very least, the model should fit the data (i.e. reproduce key patterns)
- But that is usually a low bar to clear, so additional criteria are required
- Results should also ”make sense” (i.e. conform to economic theory)
  - e.g. upward-sloping demand curve would violate this
  - a result that says agents prefer less income or profit
- Typically requires modeling heterogeneity in preferences or productivity
- Another difficulty: separating preferences from constraints or beliefs

MODEL FIT VS. IDENTIFICATION

- A model that fits the data well need not be identified
- Flexible models can match almost any set of moments
- Identification requires that:

  Different parameter values imply different distributions of observables
- Over-parameterization can weaken or destroy identification
- Good fit is *necessary*, but rarely *sufficient*

COMMON IDENTIFICATION FAILURES

- Observational equivalence
  - multiple parameter vectors imply same observables
- Weak identification
  - likelihood or moments are flat in some dimensions
- Confounding of primitives
  - preferences vs constraints
  - beliefs vs risk aversion
  - heterogeneity vs state dependence

Structural Methods

- Unlike reduced-form methods, there is not a set ”toolkit” of techniques
- Whereas RF methods almost exclusively focus on linear econometric models....
- Structural methods overwhelmingly require use of non-linear econometric models
  - e.g. discrete choice models, fixed point mappings for equilibrium models, etc.
- Structural models are typically estimated by GMM or Maximum Likelihood
- Computational know-how helps speed up the process of estimating these models
- These topics will be the focus of this class

STRUCTURAL PAPERS IN VARIOUS FIELDS

- **Labor:** Keane & Wolpin (1997), education investment decisions
- **IO:** Berry et al. (1995), demand estimation using market-level data
- **Urban:** Ahlfeldt et al. (2015), estimation of spatial agglomeration
- **Environmental:** Rudik (2020), quantify uncertainty in environmental IAMs
- **Public:** Bayer et al. (2016), dynamic Tiebout sorting model
- **Macro:** all DSGE models
- **International:** Jin & Shen (2020), coordination of sovereign debt

Internal and External Validity

- Internal validity refers to whether a causal effect in the population of interest is identified
  - often relies on specific policy variation
- External validity refers to generalizability of estimates to new contexts
- Typically, RF approaches are very good at internal validity but not at external validity
- On the other hand, if economic agents behave similarly across contexts, structural models can be externally valid
- RF and structural methods used together can improve both internal and external validity

EXAMPLE: INTERNAL VS. EXTERNAL VALIDITY

- Suppose we want to measure earth's gravitational force, g
- We can measure g by timing how long it takes various objects to fall some distance
- We can do this with objects of varying mass and of varying fall distances
  - using this data, we can estimate earth's g
- But what about the g on Mars? Or some other planet?
- For this we need a model of what exactly determines g
  - a planet's mass and proximity to other large objects
- This model will tell us what g is on planets we haven't yet visited

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